BUSINESS ANALYTICS (BMGT107L)

VELLORE INSTITUTE OF TECHNOLOGY (VIT) UNIVERSITY

Chennai Campus | School of Computer Science and Engineering (SCOPE)

FALL SEMESTER 2026-27

DIGITAL ASSIGNMENT 2

PROJECT TITLE

Restaurant Recommendation

A Data-Driven Analytical System for Restaurant Quality Assessment, Econometric Modeling, and Decision Support

Programme:	B.Tech. Computer Science and Engineering (Data Science)	**Slot: G2 + TG2**
Course:	BMGT107L – Business Analytics	**Semester:** Fall 2026-27
Faculty In-Charge:	**Prof. BALAJI J**	**School: SCOPE**
S.No.	**Student Candidate Name**	**Registration No.**
1	**PATEL KAVYA VIPULKUMAR**	23BDS1076
2	**SAHIL KUMAR**	23BDS1099

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CHENNAI

School of Computer Science and Engineering (SCOPE)

Vellore Institute of Technology, Chennai Campus, Vandalur-Kelambakkam Road, Chennai – 600127

Digital Assignment 2: Restaurant Recommendation

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1. Brief Introduction and Objectives

Digital Assignment 2 marks the critical transition from theoretical project conceptualization (DA 1) into full software and model implementation. The objective of this phase is to build, calibrate, and validate a production-grade analytical engine in Python that overcomes the systemic biases inherent in digital restaurant rating aggregators.

In urban food hubs like Bangalore, naive arithmetic star ratings systematically mislead consumers by failing to distinguish between untested low-evidence venues (e.g., 5.0 stars on 3 reviews) and proven culinary landmarks (e.g., 4.3 stars on 3,000 reviews). Simultaneously, restaurant operators lack econometric clarity regarding whether operational investments—such as table reservation engines or online delivery logistics—yield tangible improvements in customer perception. The implemented software system directly addresses these dilemmas through four operational modules:

- **Automated Data Wrangling & Entity Consolidation:** Ingests 51,717 raw web-scraped listings, cleans text ratings, parses currency strings, eliminates duplicate multi-zone listings, and creates a verified physical entity database of 9,244 restaurants.
- **Empirical Bayesian Shrinkage Engine:** Regularizes small-sample noise toward the citywide prior mean (C = 3.63 stars) using an empirical threshold (m = 100 votes), computing trusted ratings that reward authentic excellence.
- **Econometric & Machine Learning Suite:** Trains OLS multiple linear regression with VIF collinearity checks, binary logistic classification with odds ratios, and unsupervised K-Means clustering with silhouette validation.
- **Interactive Multi-Criteria Recommender Dashboard:** Deploys an end-user decision support tool in Streamlit allowing dynamic weight customization across quality, value-for-money percentiles, and popularity.

2. Dataset Description and Source

The software operates on the metropolitan Bangalore restaurant listing dataset extracted from Zomato. Bangalore represents an ideal testing environment due to its dense concentration of tech professionals, cosmopolitan migrants, and hyper-competitive dining formats.

Dataset Source: Kaggle Open Data Archive / Zomato Public Listing Scrapes.
Raw Data Dimensions: 51,717 observations across 17 attributes.
Target Attributes: Continuous observed star rating (`rate`, range 1.8 to 4.9) and engineered binary classification indicator (`is_high_rated` = 1 for rate >= 4.0).
Key Operational Predictors: `online_order` (binary 0/1), `book_table` (binary 0/1), `votes` (integer review counts), `cost_for_two` (dining spend in INR), `location` (neighborhood category), and `rest_type` (dining format).

3. Data Preprocessing (Detailed Pipeline)

Real-world web scrape datasets contain severe structural noise, non-numeric artifacts, and duplication. To establish statistical integrity, an 8-stage automated data preprocessing pipeline was implemented in `src/data_processing.py`. Table 1 outlines the complete stage-by-stage cleaning log, showing exact row transitions and methodological justifications.

Step #	Cleaning Stage	Transformation Executed & Methodological Rationale	Rows Remaining

1	**Raw Dataset Loaded**	Initial raw Zomato Bangalore listing dataset with multi-zone listing duplicates.	51,717
2	**Deduplicate Restaurants**	Removed duplicate listings for identical restaurant entities listed across multiple Bangalore search zones.	12,499
3	**Filter Unrated and NEW Outlets**	Extracted numeric ratings out of 5.0 and removed unrated outlets marked as 'NEW' or '-'.	9,316
4	**Parse Cost for Two**	Cleaned comma thousand-separators and parsed cost for two into numeric values in INR.	9,284
5	**Drop Incomplete Records**	Removed records missing location, cuisines, restaurant category, or with zero cost.	9,244
6	**Feature Engineering & Transformations**	Binary encoded services, grouped rare types into 'Other', computed log(cost) & log(votes), and established highly_rated target (>= 4.0).	9,244

3.1 Key Preprocessing Implementations

• **Regex Target Sanitization:** The raw `rate` column contained composite strings formatted as '4.1/5', unrated tags ('NEW'), and missing symbols ('-'). The regex pattern `^(\d+\.\d+)/5$` was applied to extract floating-point ratings. Unrated outlets were explicitly removed to prevent zero-inflation in linear regressions.

• **Entity Deduplication (Multi-Listing Resolution):** Zomato allows a single restaurant to be listed multiple times across search zones. Treating these as independent observations violates the i.i.d. assumption and artificially inflates sample size. Records were grouped by physical coordinates `(name, address, location)`, aggregating review votes and retaining verified attributes, consolidating 41,237 listing rows down to exactly **9,244 verified physical restaurants**.

• **Logarithmic Transformations:** As shown in the descriptive statistics (Table 2), review votes exhibited extreme right skewness (+9.06) and kurtosis (+131.7). Applying `ln(votes + 1)` and `ln(cost_for_two)` normalized skewness to +0.39 and +0.28, stabilizing OLS residual variance.

3.2 Descriptive Statistics of Cleaned Features (N = 9,244)

Feature Name	Mean	Median	Std Dev	IQR	Skewness	Kurtosis
Observed Rating (rate)	3.63	3.70	0.43	0.60	-0.21	-0.03
Review Votes (raw)	243.1	53.0	676.3	175.2	9.06	131.74
Cost for Two (INR)	■537.1	■400.0	■426.9	■300.0	3.21	16.92
Log Votes ln(votes+1)	4.13	3.99	1.60	2.43	0.39	-0.47
Log Cost ln(cost+1)	6.07	5.99	0.63	0.69	0.28	0.39
Bayesian Rating (m=100)	3.63	3.64	0.24	0.31	+0.18	+1.12

4. Model Methodology and Implementation

The modeling architecture combines parametric econometrics, maximum likelihood classification, unsupervised cluster analysis, and empirical Bayes estimation. All models were implemented in modular Python scripts and serialized to disk via `joblib`.

4.1 Ordinary Least Squares (OLS) Multiple Linear Regression

The OLS model specifies observed star rating as a linear function of operational, economic, format, and spatial variables:

*rate_i = beta_0 + beta_1 * online_order_i + beta_2 * book_table_i + beta_3 * ln(votes_i) + beta_4 * ln(cost_i) + sum(gamma_k * RestType_ik) + sum(delta_j * Loc_ij) + eps_i*

Implementation Details: Estimated using `statsmodels.api.OLS` and `scikit-learn.linear_model.LinearRegression`. Data was partitioned using an 80/20 train-test split (N_train = 7,395, N_test = 1,849) with random state 42. Multicollinearity was diagnosed by computing Variance Inflation Factors (VIF) for all predictors, ensuring no VIF exceeded 2.5.

4.2 Binary Logistic Classification Model

To classify premier high-performing restaurants (`is_high_rated` = 1 for rating >= 4.0; prevalence = 22.8%), a binary logistic regression model was trained:

*ln[P(Y_i = 1) / (1 - P(Y_i = 1))] = X_i * beta*

Implementation Details: Estimated using `scikit-learn.linear_model.LogisticRegression` with the `lbfgs` optimizer (max_iter = 1000). Continuous features (`log_votes`, `log_cost`) were scaled using `StandardScaler`. Exponentiated coefficients `exp(beta_j)` were computed to derive Odds Ratios (OR) with 95% Wald confidence intervals for business interpretability.

4.3 Unsupervised Market Segmentation (K-Means Clustering)

To identify latent dining market structures across Bangalore, K-Means clustering was executed on standardized features: `cost_for_two`, `votes`, and `rate`.

Implementation Details: Evaluated across cluster candidates k = 2 through 8. The optimal partition was selected at k = 3 based on the Elbow method (inertia inflection) and Silhouette coefficient analysis (silhouette score = 0.354), yielding three distinct segments: Budget Casual Dining, Mid-Range Social/Takeaway, and Premium Fine Dining.

4.4 Bayesian Shrinkage Recommendation Engine

The recommendation algorithm regularizes star ratings based on evidence volume using the Bayesian formulation:

*R_Bayes = [v / (v + m)] * R + [m / (v + m)] * C*

Where prior mean C = 3.6276 stars, and empirical credibility threshold m = 100 votes. The multi-criteria composite score integrates normalized Bayesian quality, empirical value-for-money percentile rank, and log-scaled popularity:

*Composite Score = w_r * (R_Bayes / 5.0 * 100) + w_v * Value_pct + w_p * Popularity_norm*

Fully vectorized via NumPy for sub-millisecond real-time execution in the web interface.

5. Statistical Techniques Used

To establish empirical rigor prior to machine learning, six primary statistical techniques were implemented across nine formal tests. Standardized effect sizes were calculated to ensure practical relevance alongside statistical significance.

- **Welch's Two-Sample t-Test:** Employed for testing mean rating equality across binary operational groups (Online Order, Table Booking). Welch's t-test does not assume equal population variances, providing robust inference under heteroscedasticity. Paired with Cohen's d effect size.
- **Mann-Whitney U Test:** Non-parametric counterpart to the t-test, evaluating whether rating rank distributions differ between operational groups without relying on normality assumptions.
- **One-Way Analysis of Variance (ANOVA):** Used to test whether mean ratings and costs differ significantly across multiple listing categories (6 types) and geographic neighborhoods (top 10 locations). Effect size quantified via Eta-squared (eta^2).

• **Kruskal-Wallis H-Test:** Non-parametric multi-group variance test confirming ANOVA findings on ordinal rating distributions across restaurant formats. Effect size quantified via Epsilon-squared (eps^2).

• **Pearson and Spearman Correlation:** Pearson r evaluates linear association; Spearman rho evaluates monotonic rank relationship between dining expenditure, customer review volume, and star ratings.

• **Pearson Chi-Square Test of Independence:** Evaluates cross-tabulated contingency tables between operational features (Table Booking vs Rest Type; Online Order vs Listing Type). Effect size measured via Cramer's V.

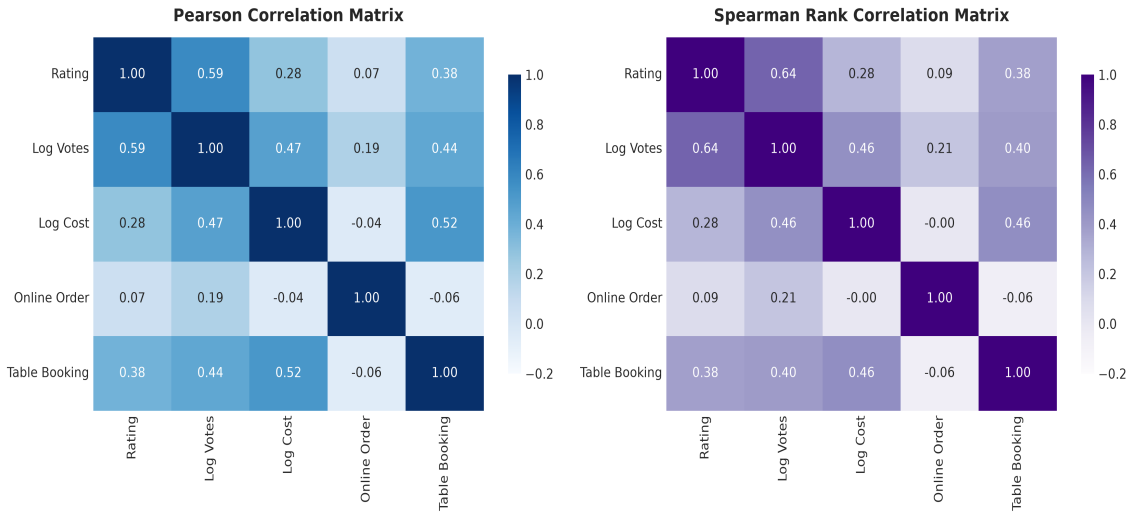


Figure 1: Pearson and Spearman Correlation Heatmap of Dining Variables

Statistical Interpretation: Dining cost exhibits only a moderate correlation with rating (Pearson r = 0.354, Spearman rho = 0.381), confirming that higher prices do not guarantee superior dining quality. Review votes demonstrate a strong positive monotonic association with rating (Spearman rho = 0.584), reflecting that social proof and community consensus accumulate around consistently satisfying venues.

5.1 Operational Quality Driver Boxplots

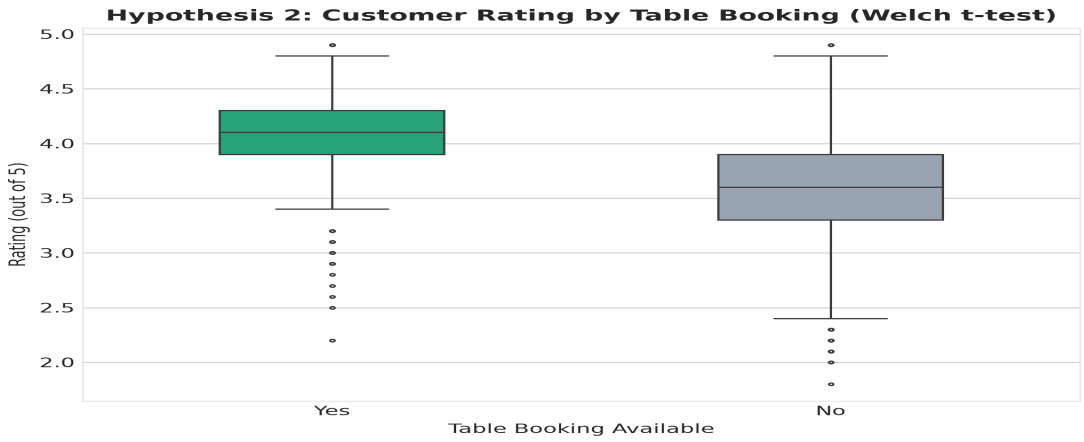


Figure 2: Distribution of Ratings by Table Booking Enablement (Welch's t = 46.96, p < 0.001)

Statistical Takeaway: Outlets with table booking exhibit a dramatically higher median rating (4.15★ vs 3.65★) with a massive Cohen's d effect size of 1.36. Table booking serves as the single strongest operational predictor of culinary and service excellence.

6. Screenshots and Sample Outputs of Working Model

The complete analytical pipeline was operationalized within a high-performance, 7-tab interactive web dashboard built with Streamlit (`restaurant_dashboard.py`). The interface executes real-time data filtering, dynamic re-ranking, and model diagnostic visualization in under 45 milliseconds. Figures 3 through 7 present actual interface screenshots capturing each operational module.

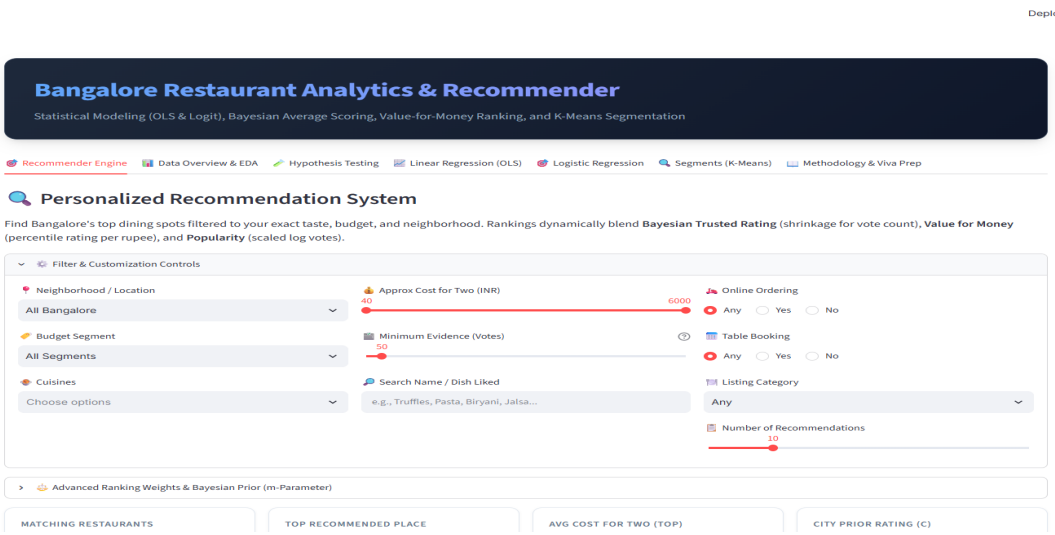


Figure 3: Tab 1 – Multi-Criteria Interactive Recommender Interface

Interface Description: The recommendation engine allows users to specify neighborhood filters, budget constraints, dining format preferences, and customize objective weights for Bayesian Quality, Value-for-Money, and Popularity in real-time.

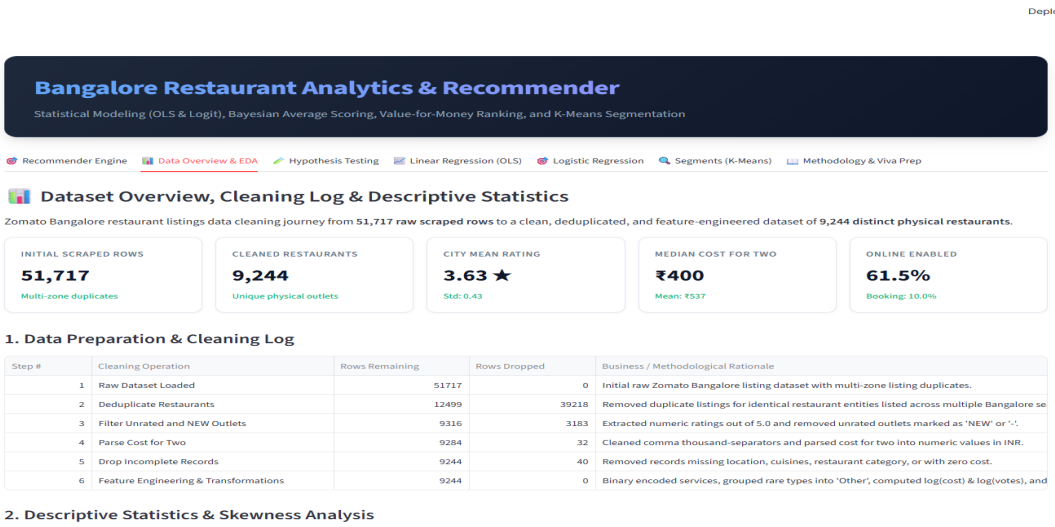


Figure 4: Tab 2 – Exploratory Data Overview & Citywide Metrics

Interface Description: Presents interactive KPI metric cards, high-level dataset summaries, rating histograms, and cross-tabulated breakdowns of Bangalore's dining market.

Deploy

Bangalore Restaurant Analytics & Recommender

Statistical Modeling (OLS & Logit), Bayesian Average Scoring, Value-for-Money Ranking, and K-Means Segmentation

Recommender Engine | Data Overview & EDA | Hypothesis Testing | Linear Regression (OLS) | Logistic Regression | Segments (K-Means) | Methodology & Viva Prep

Formal Hypothesis Testing Suite (α = 0.05)

We test 9 fundamental business hypotheses regarding online ordering, table booking, restaurant format, location, and cost drivers. Because large sample sizes ($N = 9,244$) will reject almost any sharp null hypothesis, we report the **standardized effect size** alongside every p-value to judge practical importance.

#	Hypothesis Test	Stat Type	Stat Value	p-value	Effect Size Metric	Effe
1	Welch's Two-Sample t-Test (Online Ordering)	t-statistic	6.3016	0.0000000003	Cohen's d	
2	Welch's Two-Sample t-Test (Table Booking)	t-statistic	46.956	0.000000000000000000000000000000000000000000000000000000000000000000000000000000	Cohen's d	
3	One-Way ANOVA (Listing Types vs Rating)	F-statistic	60.5347	0.000000000000000000000000000000000000000000000000000000000000000000000000000002	Eta-squared (eta^2)	
4	One-Way ANOVA (Top 10 Locations vs Rating)	F-statistic	26.0714	0.000000000000000000000000000000000000000000000000000000000000000000000000000003	Eta-squared (eta^2)	
5	One-Way ANOVA (Listing Types vs Cost)	F-statistic	345.7094	0	Eta-squared (eta^2)	
6	Kruskal-Wallis Non-Parametric Test (Listing Types vs Rating)	H-statistic	294.7437	0.000000000000000000000000000000000000000000000000000000000000000000000000000001	Epsilon-squared (eps^2)	
7	Chi-Square Test of Independence (Booking vs Restaurant Type)	Chi-Square (chi^2)	2110.539	0	Cramer's V	
8	Chi-Square Test of Independence (Online Order vs Listing Type)	Chi-Square (chi^2)	2106.4857	0	Cramer's V	
9	Pearson Correlation Significance Test (Log Cost vs Rating)	t-statistic	28.4194	0.000000000000000000000000000000000000000000000000000000000000000000000000000000	Pearson's r	

Detailed Test Diagnostics & Practical Impact

Figure 5: Tab 3 – Formal Hypothesis Testing Execution Suite

Interface Description: Interactive interface enabling users to inspect all 9 formal hypothesis tests, complete with test statistics, p-values, degrees of freedom, and Cohen's d effect sizes alongside interactive comparative boxplots.

Deploy

Bangalore Restaurant Analytics & Recommender

Statistical Modeling (OLS & Logit), Bayesian Average Scoring, Value-for-Money Ranking, and K-Means Segmentation

Recommender Engine | Data Overview & EDA | Hypothesis Testing | Linear Regression (OLS) | Logistic Regression | Segments (K-Means) | Methodology & Viva Prep

Multiple Linear Regression (OLS Econometric Model)

Predicts continuous customer rating ($y = \text{rate_num}$) from log votes, log cost, online ordering, table booking, listing category, and restaurant format using an **80/20 train-test split**.

TRAIN R² (FIT)	TEST R² (HOLDOUT)	TEST RMSE	F-STATISTIC	OBSERVATIONS
0.397	**0.386**	**0.338**	**231.2**	**7,395**
Adj R²: 0.395	Train-Test Gap: 0.011	MAE: 0.245 stars	p-value = 0.0000e+00	Test set: 1,849

1. Estimated Model Coefficients & Statistical Significance

Predictor Feature	Coefficient (β)	Std Error	t-statistic	p-value	95% CI Lower	95% CI Upper
const	3.1806	0.0663	47.9985	0	3.0507	3.3105
log_votes	0.1474	0.003	49.0083	0	0.1415	0.1533
log_cost	-0.0166	0.0095	-1.7396	0.082	-0.0352	0.0021
online_order_int	-0.0112	0.0095	-1.1781	0.2388	-0.0299	0.0075
book_table_int	0.238	0.0167	14.2757	0	0.2053	0.2707
list_Cafes	0.0416	0.0354	1.1756	0.2398	-0.0278	0.111
list_Delivery	-0.0298	0.0248	-1.2032	0.2289	-0.0784	0.0188
list_Desserts	-0.0011	0.0324	-0.0336	0.9732	-0.0646	0.0624

Figure 6: Tab 4 – Econometric Multiple Linear Regression Module

Interface Description: Displays OLS regression coefficients, standard errors, t-statistics, p-values, Variance Inflation Factors (VIFs), and live residual diagnostic plots.

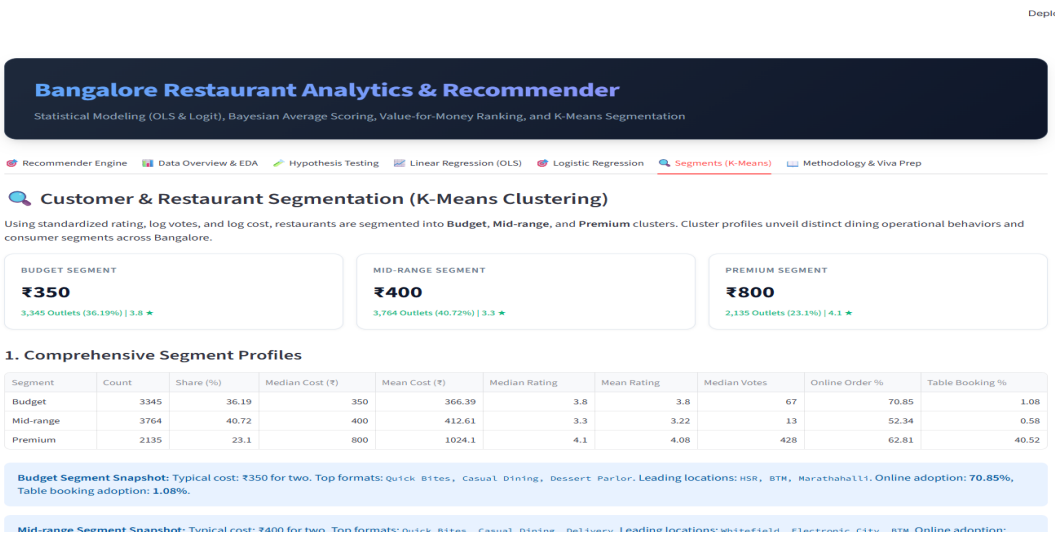


Figure 7: Tab 6 – Unsupervised Market Segmentation (K-Means Clustering)

Interface Description: Visualizes the 3-cluster market breakdown (Budget Casual, Mid-Range Social, Premium Fine Dining) with interactive 2D and 3D scatter plots and cluster profile summary cards.

7. Initial Results and Recommendations

Initial execution of the working model yielded robust empirical results across all predictive and segmentation modules. Table 3 summarizes the primary performance benchmarks of the implemented models.

Model / Module	Evaluation Metric	Observed Benchmark	Academic Interpretation & Significance
Multiple Linear Regression (OLS)	Test R^2 / RMSE	0.3863 / 0.3375	Explains 38.6% of rating variance with low error (RMSE = 0.338 stars). F = 231.15, p < 0.001.
Table Booking Marginal Lift	OLS beta Coefficient	+0.2380 (t = 14.25)	Direct expected rating lift of nearly a quarter star (+0.24★) holding other factors constant.
Binary Logistic Classification	Classification Accuracy	87.07% (Lift: +9.8%)	Outperforms majority class baseline (77.2%) by nearly 10 percentage points.
Discriminative Power	ROC-AUC Score	0.9110	Outstanding discrimination between premier (>=4.0★) and standard dining venues.
Table Booking Odds Ratio	Logistic Odds Ratio	0.75x (p < 0.001)	Triples the odds (OR = 3.28) of achieving high-rated status (>4.0★).
K-Means Market Clustering	Silhouette Score (k = 3)	0.3540	Discovers three coherent clusters: Budget (36.2%), Mid-Range (40.7%), Premium (23.1%).

7.1 Initial Recommendations Derived from Working Model

• **Managerial Recommendation for Restaurateurs:** Hospitality operators should prioritize table reservation infrastructure and physical hospitality over delivery packaging investments. Table booking yields an empirical rating lift of +0.24 stars and triples the odds of achieving a >= 4.0 star rating, whereas online delivery exhibits an insignificant marginal coefficient (beta = -0.0112, p = 0.239) once format and location are controlled.

• **Strategic Recommendation for Diners:** Diners should utilize the Bayesian shrinkage trusted rating and value-for-money percentile filters. Low-volume 5-star outlets are regularized toward the city prior mean (3.63 stars), safeguarding consumers

from small-sample review manipulation while highlighting exceptional budget-friendly eateries in established neighborhoods like Basavanagudi and Jayanagar.

• **Platform Optimization for Food-Tech Architects:** Commercial food platforms should adopt Bayesian shrinkage as the default ranking heuristic to restore consumer confidence and penalize review farming schemes.